Capgemini

Honesty
Boldness Trust
Freedom
Team Spirit
Modesty
Fun

Nozhan Balafkan

Data Scientist & Application Consultant

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Nozhan is a data scientist with a focus on integration of computer vision and machine learning in remote sensing and geospatial analysis using a wide range of optical and radar satellite imageries. He has good knowledge and experience of using technologies, frameworks and methods such as Python, Fortran, C++, deep learning, machine learning, computer vision, ArcGIS, Qgis, GitHub, GitLab, Docker, SQL databases, embedded computing boards, Satellite Imagery, Remote Sensing Analysis, GIS, Cartography, Data visualisation.

Nozhan has experience of analysing a diverse range of datasets from space physics to climate physics to energy. He is active, hard working, ambitious, collaborative, an avid learner, a good communicator and constantly seeks to improve his skills. He is a team player and cares about his team members and he is actively seeking to find the opportunity to apply his computer and problem solving skills to real world problems to provide insightful solutions.

Engagement

Remote sensing-based subsidence and rise detection

Project Description: The project aimed to study the feasibility of using interferometric synthetic-aperture radar to measure subsidence and rise in the city of Oslo over a period of one year.

Role and Responsibility: Nozhan developed the entire work flow from data collection to data analysis to mapping and visualisation. He used a persistent scatterer interferometer approach to detect subsidence and rise in different parts of the city of Oslo.

Methods and Softwares: For the purpose of this project Synthetic Aperture Radar (SAR) data of the Sentinel 1 was used. Also, softwares and programming languages such as Sentinel-1 tool box, Mapbox, Matlab, HTML, Javascript were used.

Satellite based river flow forecasting; Navarra technologies

Project Description: The project aimed to study the feasibility of using remote sensing analysis, satellite imagery, and artificial intelligence to predict the river flow of several rivers.

Role and Responsibility: Nozhan was part of a team where he used his skills in remote sensing, satellite imagery, and image processing to provide river flow prediction. As an important part of the project satellite coverage of the rivers of interest was analysed for the possible temporal and spatial resolution.

Methods and Softwares: For the purpose of this project satellite images of the Sentinel 2, Envisat, Moderate Resolution Imaging Spectroradiometer (MODIS), and Synthetic Aperture Radar (SAR) data were used. Also, softwares and programming languages such as ArcGIS, QGIS, Brat, and python were used.

AI based safety system; Capgemini

Project Description: The project goal was to provide the possibility of a computer vision based solution to help motor cyclists to detect an emergency car (i.e police car) earlier from behind. This way a motor cyclists will have more time to react to the situation appropriately.

Role and Responsibility: Nozhan developed the solution using a Jetson nano developer kit and YOLO (You Only Look Once) a deep learning Convolutional Neural Network (CNN) based model. The system is able to detect police cars behind a motorcycle via a camera mounted on the back of the motorcycle. Once a police car is detected Jetson will send a signal to an LED located in front of the motorcycle driver.

Methods and Softwares: Python, YOLO deep learning model, TensorFlow, Keras, GitHub, Logitech HD Pro C920 Web camera.

Computer vision based robot arm; Capgemini

Project Description: The goal was to enable a robotic arm to see, detect, and finally move the arm towards a specific object and grab it. In order to achieve this goal, the project was divided into two parts; the computer vision part and robotic arm movement.

Role and Responsibility: In order to accomplish the object detection part of the project Nozhan used both YOLO and RetinaNet computer vision models and these models were run using Ubuntu operating system. Finally, to enable the robot arm to see, a Logitech HD Pro C920 web camera was mounted on the robotic arm. After accomplishing the computer vision model, Nozhan contributed to the robotic arm movement part by running the Kinetic distribution of the Robotic Operating System (ROS).

Methods and Softwares: Niryo One robotic arm, Python, ROS, YOLO and RetinaNet computer vision models, TensorFlow, Keras, Rasbeppery Pi computer V3, GitHub, Logitech HD Pro C920 Web camera.

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AI based land cover classification; Capgemini

Project Description: The goal was to develop a machine learning based model that is capable of taking in satellite imageries as inputs, processing them, detecting different land cover types, and finally returns the processed images back to the user.

Role and Responsibility: A machine learning model was trained. Next, the model was deployed into a Docker container on a local server. Also, an HTML page with load functionality was design. The website was connected to the Docker container were the machine learning model takes the image(s) and processes them. The connection between the website and the Docker container was established using Flask.

Methods and Softwares: Scikit-learn, Flask, GitHub, Docker, HTML, Sentinel-2 satellite imagery.

Sensor data analysis and visualisation; Smarte Byre Norge

Project Description: The main goal was to analyse and visualise sensor-collected air pollution data of an air quality monitoring station at the Våland district in Stavanger.

Role and Responsibility: Nozhan created a time series of three air pollutants (i.e. CO2, CO, NO) along with animating the pollution variation over the Våland district in Stavanger. The results were presented at the Nordic Edge conference in Stavanger in September 2018.

Methods and Softwares: Python, GeoPandas, Matplotlib, GeoJson, Mapbox

Weather Forecast Improvement; Iran National Meteorological Organisation

Project Description: The goal was to study the feasibility of integration of machine learning and deep learning models for the purpose of improving weather prediction.

Role and Responsibility: Nozhan's work spanned from extensive literature review to analysing WRF output data, time series analysis, and training deep learning and neural networks models.

Methods and Softwares: Python, GeoPandas, Matplotlib, Neural Network, deep learning

Emission Inventory Development; Air Quality Control Company

Project Description: The goal was to develop and digitalise the emission inventory of the railway fleet of the city of Tehran.

Role and Responsibility: Nozhan's role was twofold. First, to expand the emission inventory of the railway fleet of Tehran by means of extensive literature review and field interview. Second, to digitalise the existing emission inventory.

Methods and Softwares: Python, Pandas, Matplotlib, SQL

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Education

2016 - 2018 **Master in Climate Physics and Energy**
University of Tromsø, Norway

2014 - 2015 **Graduate Researcher in Computational Astrophysics**
Saint Mary's University, Canada

2011 - 2013 **Master in Astrophysics**
University of Leipzig, Germany

Employment

2019.02 - 2020.07 **Capgemini**
Data Scientist and Application Consultant

2018 - 2019 **Iran National Meteorological Organisation**
Researcher

Air Quality Control Company (AQCC)
Data analyst

Smarte Byre Norge
Data Analyst and Data Visualiser

Certification

- AWS Fundamentals: Going Cloud-Native Amazon Web Services; Coursera
- GIS Imagery, Automation, and Applications; Coursera
- GIS and Environmental Analysis; Coursera
- GIS Data Formats, Design and Quality; Coursera
- Fundamentals of GIS; Coursera
- Convolutional Neural Networks; Coursera
- Structuring Machine Learning Projects; Coursera
- Neural Networks and Deep Learning; Coursera
- Improving Deep Neural Networks: Hyperprameter tuning, Regularisation and Optimisation; Coursera
- Model Output Statistics (MOS) Methods, Applications, and Verification; Iran National Meteorological Organisation
- Advances in Regional and Global Atmosphere – Ocean Modelling; Iran National Meteorological Organisation

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- Kenda: Ensemble Data Assimilation in Regional Model; Iran National Meteorological Organisation
- Data Retrieval for Data Assimilation; Iran National Meteorological Organisation

Language

English: Fluent

Norwegian: Basic

German: Elementary

Persian: Native

Software

Python
Fortran
C++
Javascript
HTML
CSS
Git
Machine Learning
Deep Learning (TensorFlow)
Cloud-Native Amazon Web Services (AWS)
Sentinel-1 toolbox
Sentinel-2 toolbox
PostgreSQL
Linux
Windows
MacOS
Bash
ArcGIS
QGIS
Grass
Docker
Robotic Operating System (ROS)

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